**Programming 1: Lab 7: Loops**

1. Find the sum of the following series for N terms, where N is taken as input from the user and display the result till exactly 9 decimal places.
   (a) 1 + 1/3 + 1/5 + 1/7 + 1/9 + ….. N terms.
   (b) 1/2! + 1/4! + 1/6! + … + 1/(2N)!
   (c) $1 - 1/2^3 + 1/3^3 - 1/4^3 + \ldots.$ + N terms
   (d) $1 + x^2/2! + x^4/4! + x^6/6! + \ldots.$ + N terms (Take x as input from user)
   (e) $F(x) = x/1 + x^2/3 + x^3/5 + x^4/7 + \ldots..$ + N terms (Take x also as input)

2. Find the sum of the series where x and a positive error ϵ is taken as input from the user. The value of F(x) is displayed along with the minimum number of terms that gives the valid result such that $| F_i(x) - F_{i-1}(x) | < \epsilon$ where i is the term count
   (a) $F = \Sigma_n ( 1 / 3 )^n$ where n is the term count
   (b) $F(x) = 1 + x/1! + x^2/2! + x^3/3! + \ldots..$
   (c) $F(x) = x - x^2/2 + x^3/3 - x^4/4 + \ldots..$

3. **Lucky Bus Tickets.** The tickets of a bus are numbered from x to y. A ticket is called lucky if its number reads the same forwards and backwards, for example 99 or 121. Find the sum of the numbers of all the lucky tickets of the bus, both x and y included.

   **Input:** Two integers x and y, each on a separate line.
   **Output:** A single integer - the sum of all the lucky ticket numbers.
   **Constraints:** 1 ≤ x ≤ y ≤ 100000.

| Sample Input | Sample Output |
|---|---|
| 95<br>125 | 432 |
| 30<br>44 | 77 |

   *In the first case the lucky tickets are 99, 101, 111 and 121.*

4. Print the following patterns (take number of lines as input from user):

   (a)
   ```
   ABCD
   ABC
   AB
   A
   ```

   (b)
   ```
      A
    A B
  A B C
A B C D
   ```

   (c)
   ```
A
B B
C C C
D D D D
   ```

   (d)
   ```
* * * * *
*       *
*       *
* * * * *
   ```

5. Print the Pascal's triangle for N rows, where N is taken as input from the user. Do not use any inbuilt function for factorial or combinations.

```
    1
   1 1
  1 2 1
 1 3 3 1
1 4 6 4 1
(till N rows)
```

6. Approximate the value of π using the series π = 4 ( 1 – 1/3 + 1/5 – 1/7 + ….. ) for N terms, where N is taken as input from the user. Display the result till exactly 9 decimal places along with the absolute difference from the value of math.pi.

7. Display the following pattern for N lines, where N is taken as input from the user:

```
    1
   121
  12321
1234321
(till N lines)
```

8. **Saving for a Laptop.** A student saves S rupees in the first month and then increases the amount saved by D rupees in every following month. That is, the savings are S, S + D, S + 2D, and so on. Find the month in which the total savings become at least equal to the price T of the laptop, and the total savings in that month.

   **Input:** Three integers S, D and T, each on a separate line.
   **Output:** Print the month number in the first line and the total savings in that month in the second line.
   **Constraints:** 1 ≤ S ≤ 10000, 0 ≤ D ≤ 10000, 1 ≤ T ≤ 10000000.

| Sample Input | Sample Output |
|---|---|
| 500<br>100<br>3000 | 5<br>3500 |
| 1000<br>0<br>2500 | 3<br>3000 |

   *In the first case the totals are 500, 1100, 1800, 2600 and 3500.*

9. Find the sum of the series 1 + 11 + 111 + 1111 + ….. for N terms, where N is taken as input from the user. Do not use any string operations.

10. Take a positive integer N as input and display all the numbers between 1 and N that are equal to the sum of the factorials of their own digits. For example, 145 = 1! + 4! + 5!. Do not use any inbuilt factorial function.